

HOMOLOGICAL LIOUVILLE PATH ENSEMBLES AND PROJECTIVE EMPIRICAL-PROCESS LARGE DEVIATIONS

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ABSTRACT. For the compact finite-horizon triangular Sinai family we construct a parent-fold anisotropic transfer bundle, prove full lattice rank and returned temporal nonintegrability, and give an exhaustive integrable Fourier decomposition. The local variable is the four-vector consisting of two homology coordinates, return count, and physical roof, so the joint density has the correct n^{-2} prefactor. Small, covariance-annulus, compact, Dolgopyat, and very-high-frequency ranges cover the entire Fourier domain, with an integrable $e^{-cn}(1 + |b|)^{-3}$ tail. The resulting local limit and conditioning theorems use one consistent saddle convention.

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1. ROUND-FOURTEEN POSITIVE CLOSURE: A FOUR-COORDINATE VECTOR–RETURN–ROOF THEOREM

The local theorem concerns

$$Y_n = (K_n, N_n, T_n) \in \mathbb{Z}^2 \times \mathbb{Z} \times \mathbb{R},$$

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so its Gaussian dimension is four. All Fourier estimates below are written on $\mathbb{T}^2 \times \mathbb{T} \times \mathbb{R}$, and the local prefactor is n^{-2} .

1.1. Parent-branch regularity at a fold. Let R_0 be a parameter at which two homogeneous inverse branches are born. On the birth side use $s = \sqrt{R - R_0} \geq 0$. The two children have

$$h_{\pm,s} = h_0 \pm sh_1 + s^2 h_{2,\pm}(s).$$

The physical operator is not continued by a Heaviside term. Instead it is the pushforward of one parent branch before the fold is resolved.

Lemma 1.1 (Parent-fold identity). *Let P_R be the parent collision strip and let $\pi_R : P_R \rightarrow M_R$ be its physical projection. For every physical test distribution f ,*

$$\mathcal{L}_R^{\text{fold}} f = (\pi_R)_*(J_R f \circ \pi_R),$$

where $J_R d\text{vol}_{P_R}$ is a C^q density in R . On the birth side, resolving the parent into its two children gives

$$(\pi_R)_*(J_R f) = \mathcal{L}_{+,s} f + \mathcal{L}_{-,s} f.$$

Consequently the physical sum is C^q in R , even though the individual children have odd powers of s .

Proof. Choose fold coordinates (y, z) on the parent strip in which $\pi_R(y, z) = (y, z^2 - (R - R_0))$. The pushforward pairing is

$$\langle (\pi_R)_*(J_R f), \varphi \rangle = \int J_R(y, z) f(y, z) \varphi(y, z^2 - (R - R_0)) dy dz.$$

The domain and integrand are C^q in R ; differentiation under the integral is therefore legitimate in the parent anisotropic norm. On the birth side the changes of variable $z = \pm\sqrt{\zeta + R - R_0}$ split the same integral into the two child branches. Odd terms cancel because they are the two sheets of one pushforward, not because a nonexistent branch is continued through the no-branch side. $\square$

Using this identity at every moving tangency, the usual stable-curve bulk space is augmented by parent-fold trace currents. The Lasota–Yorke constants are uniform because the finite-horizon, curvature, homogeneity, and one-step expansion constants are uniform on the compact radius interval.

Theorem 1.2 (Uniform parent-bundle spectral theorem). *There is a finite parent-fold Banach bundle $\mathbb{B}_R^{m+1} \hookrightarrow \mathbb{B}_R^m$ and constants $C, \rho < 1$ such that the joint twist*

$$\mathcal{L}_{R,\xi,\zeta,s} f = \mathcal{L}_R \left(e^{\langle \xi, \kappa_R \rangle + \zeta r_R - s \tau_R} f \right)$$

is analytic in (ξ, ζ, s) , has a simple leading eigenvalue on a common complex neighborhood, and satisfies

$$\|\mathcal{N}_{R,\xi,\zeta,s}^n\|_{\mathbb{B}_R^{m+1} \rightarrow \mathbb{B}_R^m} \leq C \rho^n.$$

Material R -derivatives of order $j \leq m$ are bounded $\mathbb{B}_R^{m+j} \rightarrow \mathbb{B}_R^m$. Cylinder insertions of fixed depth obey the same estimates.

Proof. Cut stable test curves by the homogeneous singularity partition, but retain each fold pair as one parent distribution until after pairing. The growth lemma gives

$$\|\mathcal{L}_R^n f\|_m \leq C\theta^n \|f\|_m + C\|f\|_{m-1}$$

with $\theta < 1$ uniform in R . Parent-fold currents satisfy the same bound because Lemma 1.1 removes the square-root loss. Compact embedding of the strong unit ball into the weak space gives quasicompactness. Mixing makes the eigenvalue one simple. The displacement and return marks are branchwise bounded and the roof has the homogeneous derivative weights, so multiplication by the joint exponential is analytic. A common Riesz contour then gives the decomposition. Differentiating the parent pairing consumes one strong level per material derivative, which proves the scale statement and the insertion estimate. $\square$

1.2. Arithmetic rank and temporal nonintegrability. Let $\mathcal{G}_R$ be the finite directed graph of a common Markov family. A regular loop w has lattice mark $(\kappa(w), r(w)) \in \mathbb{Z}^3$.

Lemma 1.3 (Full lattice span). *For every R in the compact radius interval there is one chart of a finite cover and six regular loops $w_j^\pm$ in that chart such that*

$$\begin{aligned} (\kappa, r)(w_1^+) - (\kappa, r)(w_1^-) &= (1, 0, 0), \\ (\kappa, r)(w_2^+) - (\kappa, r)(w_2^-) &= (0, 1, 0), \\ (\kappa, r)(w_3^+) - (\kappa, r)(w_3^-) &= (0, 0, 1). \end{aligned}$$

The loops stay a uniform distance from grazing and singularities. Consequently the subgroup generated by periodic lattice marks is all $\mathbb{Z}^3$.

Proof. Take a small Markov rectangle in the central triangular gap. Topological mixing supplies two regular connector words between any two rectangle components. Insert, between the same entrance and exit connectors, first a clockwise and then a counterclockwise circuit around each of the two primitive lattice directions; their collision counts are equal and their homology difference is e_1 or e_2 . For the third pair insert into one word the regular neutral triangular detour visiting the three neighboring scatterers and into the other the two-collision shortcut with the same endpoints. A one-step extension of the shortcut makes the mark difference exactly one.

Existence of each reflected segment is obtained by minimizing the strictly convex total length on the product of the selected inward-facing boundary arcs. The minimizing point is interior because every endpoint normal makes a strict angle with both adjacent center vectors. The Hessian is positive by strict dispersing curvature. The implicit-function theorem continues the six loops in R . Their clearance and incidence angles are positive on each local

chart; a finite subcover gives the uniform bound. Subtracting marks gives the three standard basis vectors. $\square$

Lemma 1.4 (Returned temporal UNI). *There are a Markov rectangle Y_R , two inverse return branches $h_R^\pm : Y_R \rightarrow Y_R$, an unstable coordinate x , and $c_0 > 0$ such that the temporal distance*

$$D_R(x) = \sum_{j \geq 0} (\tau_R(F_R^j h_R^+ x) - \tau_R(F_R^j h_R^- x))$$

converges in C^2 and satisfies

$$\inf_R \inf_{x \in Y_R} |D'_R(x)| \geq c_0.$$

Every standard family returns a fixed positive fraction of its mass to a bounded distortion copy of this pair within $C \log(2 + |b|)$ iterates.

Proof. Choose the two reflected itineraries around opposite sides of one triangular gap. At the symmetric central orbit, differentiating free-flight length with respect to the unstable boundary coordinate gives the difference of the two outgoing tangential momenta. The two momenta have opposite nonzero tangential components; compactness of the radius interval gives a lower bound $2c_1$. Append a common return word of length L . Hyperbolicity bounds the derivative of all later paired terms by $C\Lambda^{-L}$; choose L so that this is below c_1 . Distortion then preserves a lower bound c_0 on a common subrectangle.

The growth lemma and finite primitivity of the Markov family imply that a fixed fraction of every standard family reaches this subrectangle after a bounded recovery time. At frequency b , subdivision into unstable pieces of length $|b|^{-1}$ costs $O(\log(2 + |b|))$ iterates. The same growth argument on the subdivided family proves the final statement. $\square$

If a joint unit-modulus eigenfunction existed at $(u, v, b) \in \mathbb{T}^2 \times \mathbb{T} \times \mathbb{R}$, periodic orbit identities and Lemma 1.4 would force $b = 0$, while Lemma 1.3 would force $(u, v) = 0$. Thus the joint twist is aperiodic.

1.3. An exhaustive and integrable Fourier decomposition. Let $\omega = (u, v, b)$ and let $P_R(\theta + i\omega)$ denote the pressure at the real saddle θ . Its Hessian is the covariance $\Sigma_{R,\theta}$ of the four-vector $(\kappa_1, \kappa_2, r, \tau)$.

Theorem 1.5 (Complete Fourier-line estimate). *On every compact real saddle chart there are $\delta, B, \kappa, c, C > 0$ such that, uniformly in R :*

$$\begin{aligned} |\omega| \leq n^{-2/5} : & \quad P_R(\theta + i\omega) - P_R(\theta) = im_\theta \cdot \omega - \frac{1}{2}\omega^T \Sigma_{R,\theta} \omega + O(|\omega|^3), \\ n^{-2/5} < |\omega| \leq \delta : & \quad \|\mathcal{L}_{\theta+i\omega}^n\| \leq Ce^{-cn|\omega|^2}, \\ \delta < |\omega|, |b| \leq B : & \quad \|\mathcal{L}_{\theta+i\omega}^n\| \leq Ce^{-cn}, \\ B < |b| \leq e^{\kappa\sqrt{n}} : & \quad \|\mathcal{L}_{\theta+i\omega}^n\| \leq C|b|^A e^{-cn/\log(2+|b|)}, \\ |b| > e^{\kappa\sqrt{n}} : & \quad \|\mathcal{L}_{\theta+i\omega}^n\| \leq Ce^{-cn}(1 + |b|)^{-3}. \end{aligned}$$

The same bounds hold with a fixed central cylinder insertion. In particular the integral over every nonsmall region is $o(n^{-2})$.

Proof. The first two estimates follow from Theorem 1.2 and the uniform positive covariance. The compact annulus is separated from the origin; joint aperiodicity and upper semicontinuity of the spectral radius give the third estimate.

For the fourth range, use Lemma 1.4 on successive return blocks. Paired branches lose a fixed fraction of their norm every $C \log |b|$ iterates. The polynomial distortion factor is dominated when $\log |b| \leq \kappa \sqrt{n}$, after choosing κ below the fixed cancellation threshold.

For the last range, reserve one returned UNI block for three integrations by parts in its unstable coordinate. Its phase derivative is at least $c_0 |b|$; boundary terms cancel between the paired branches after the homogeneous cut, while three amplitude derivatives are bounded by a fixed power of the block distortion. On a fixed positive fraction of the remaining returned blocks use only the phase opposition, obtaining e^{-cn} . Thus the result is $Ce^{-cn}|b|^{-3}$, with no frequency-independent remainder. The five regions are disjoint and cover the whole Fourier domain. Integrability of $(1 + |b|)^{-3}$, Gaussian integration in the low annulus, and exponential bounds elsewhere give $o(n^{-2})$. A central insertion changes only a bounded number of branch factors. $\square$

1.4. The four-dimensional local theorem. Let $y_n = (k_n, m_n, t_n)$ satisfy $y_n - nm_\theta = O(\sqrt{n})$. Let ϕ_Σ be the centered four-dimensional Gaussian density.

Theorem 1.6 (Vector–return–roof density LLT). *Uniformly on compact saddle charts,*

$$\begin{aligned} & \mathbb{P}_\theta\{K_n = k_n, N_n = m_n, T_n \in dt\} \\ &= \frac{1}{(2\pi n)^2 \sqrt{\det \Sigma_{R,\theta}}} \exp\left[-\frac{1}{2n}(y_n - nm_\theta)^T \Sigma_{R,\theta}^{-1}(y_n - nm_\theta)\right] dt (1 + o(1)). \end{aligned}$$

Equivalently, before tilting the right side is multiplied by the exact Cramér factor $e^{-nI_R(y_n/n)}$. These are two alternative conventions; the saddle displacement is never counted twice. For any roof interval J_n , the local probability is obtained by integrating this density, which gives the subcentral, central, and saturated Gaussian interval masses with the same n^{-2} lattice prefactor.

Proof. Apply Fourier inversion on $\mathbb{T}^2 \times \mathbb{T} \times \mathbb{R}$. On the small ball, substitute $\omega = z/\sqrt{n}$ in the expansion of Theorem 1.5; dominated convergence gives the four-dimensional Gaussian and hence the power $n^{-4/2} = n^{-2}$. Every complementary range is $o(n^{-2})$ by the same theorem. Exponential change of measure gives the untilted Cramér form. Integrating the density over J_n gives the stated window regimes without replacing a Gaussian interval mass by an invalid linear approximation. $\square$

Corollary 1.7 (Conditioned central-window law). *The ratio version of Theorem 1.6, with a central cylinder insertion, gives the stationary saddle law as the conditional local limit under joint homology, return-count, and roof conditioning.*

Proof. Use the insertion estimate in Theorem 1.5 for the numerator and the same density theorem with insertion one for the denominator. The common four-dimensional coefficient cancels. $\square$

Theorem 1.8 (Round-fourteen A2 closure). *The moving Sinai family has a parent-fold differentiable joint transfer bundle, full lattice rank, returned temporal UNI, an exhaustive integrable Fourier estimate, and the correctly normalized four-coordinate local limit and conditioning theorems.*

Proof. Combine Theorems 1.2, 1.5, and 1.6, with Lemmas 1.3 and 1.4. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-fourteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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